QUANSER
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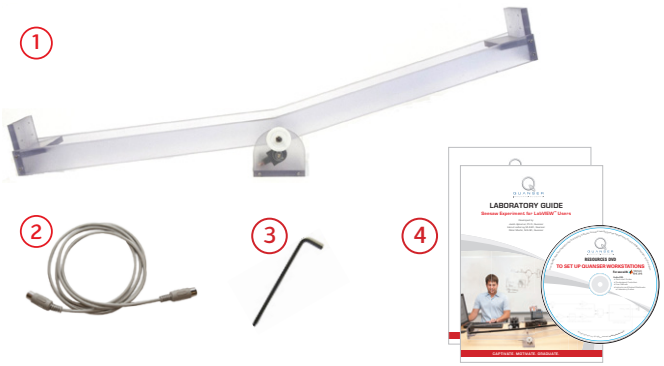
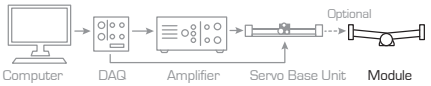
Quick Start Guide: Seesaw

MATLAB SIMULINK

STEP 1 Check Components and Details

Make sure your Seesaw module includes the following components:



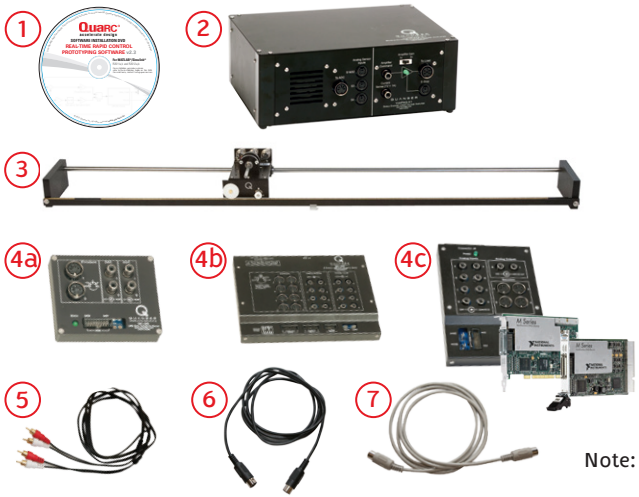


1. Seesaw module
2. 5-pin DIN to 5-pin DIN encoder cable
3. 7/64' Allen key
4. Quanser Workstations Resources DVD* (includes controllers; digital versions of User Manual, Quick Start Guide and courseware; and other files)

*DVD supplied with the QUARC Real-Time Rapid Control Prototyping Software, see Step 2

STEP 2 Additional Components Required for Set Up

To complete the Seesaw set up, you will also need the following:



1. QUARC Real-Time Rapid Control Prototyping Software Installation DVD
2. Power Amplifier (VoltPAQ-X1 pictured)
3. Linear Servo Base Unit
4. One of the following data acquisition devices:
   a. Quanser Q2-USB, or
   b. Q8-USB, or
   c. NI PCI/PCIe with Terminal Board
5. RCA to RCA cable
6. 4-pin DIN to 6-pin DIN motor cable
7. 5-pin DIN to 5-pin DIN encoder cable

Note: These components must be purchased separately.

STEP 3 Install and Test QUARC

A. Make sure you have all required software, as listed in the QUARC Compatibility Table document located in the QUARC DVD folder.

B. See the QUARC Installation Manual for details on how to install the software.

C. Make sure you test the system using the Sine and Scope demo. You can access this by typing **qc_show_demos** in the Matlab prompt.

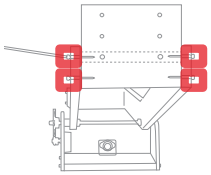
STEP 4 Set Up the Hardware

To set up your Seesaw module, please read the following instructions carefully. The connections shown below are illustrated using a generic data acquisition (DAQ) device and a VoltPAQ-X1 amplifier (you may have a different DAQ or amplifier). For detailed instructions, see the Seesaw User Manual (enclosed with shipment).

A Before proceeding, set up and test your Linear Servo Base Unit. For detailed instructions, see the Linear Servo Base Unit Quick Start Guide or User Manual.

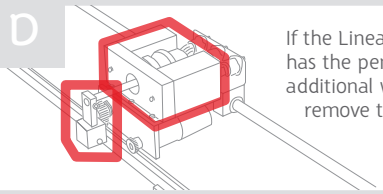
B Make sure everything is powered OFF before making any connections. This includes turning off your PC and the amplifier.

C



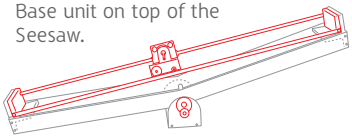
Loosen the four screws on both ends of the Seesaw.

D



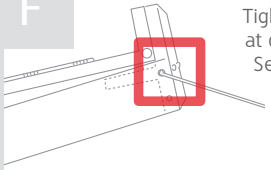
If the Linear Servo Base Unit has the pendulum rod or the additional weight mounted, remove them.

E Place the Linear Servo Base unit on top of the Seesaw.



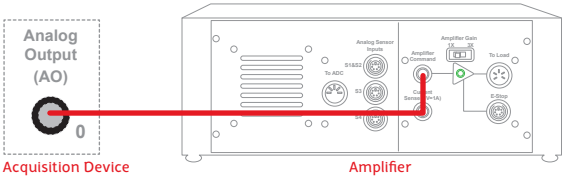
Note: Make sure the Linear Servo Base cart pinions are on the same side as the Seesaw gear.

F



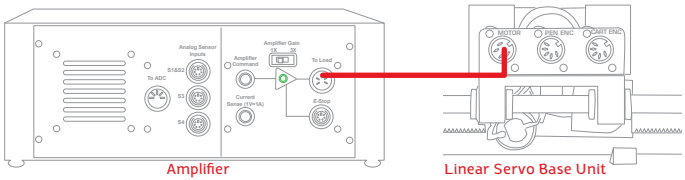
Tighten the four screws at on each end of the Seesaw to secure the Linear Servo Base from sliding.

G



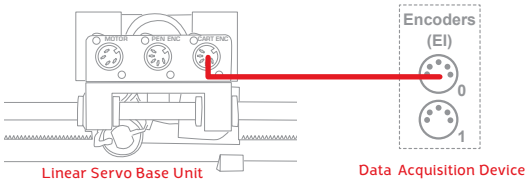
Using the RCA to RCA cable, **connect Analog Output Channel** #0 (AO #0) on the data acquisition (DAQ) device to the **Amplifier Command** socket on the amplifier.

H



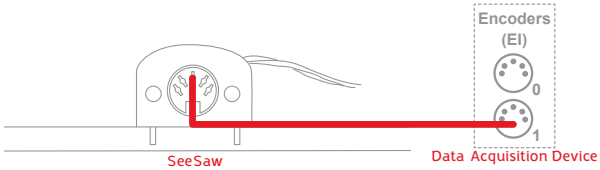
Using the 4-pin DIN to 6-pin DIN motor motor cable, connect the **To Load** socket on the amplifier to the **Motor** socket on the Linear Servo Base Unit.

I



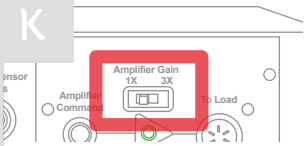
Using the 5-pin DIN to 5-pin DIN encoder cable, connect the **Encoder Channel #0** (EI #0) on the data acquisition (DAQ) device to the **Cart Encoder** connector (CART ENC) on the Linear Base Unit.

J



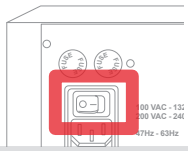
Using a 5-pin DIN to 5-pin DIN encoder cable, connect the **Encoder Channel #1** (EI #1) on the data acquisition (DAQ) device to the **Encoder** connector on the Seesaw.

K



Attention VoltPAQ-X1 Users: Make sure you set the GAIN on the VoltPAQ-X1 to 1 when using any Linear Servo Base Unit experiment.

L



Turn ON the power switch on the VoltPAQ-X1. It is located on the rear of the device.

STEP 5 Testing the Seesaw

Follow the procedure below to test your Seesaw module.

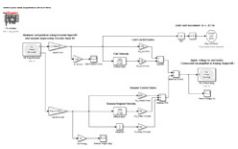
A

Make sure your PC and amplifier are powered ON.

B

On the Resources DVD (supplied with the QUARC and Linear Base Unit package), locate the **Quick Start Folder**: Rotary\Rotary Servo Base Unit\Quick Start. Copy the Quick Start folder to your local hard drive.

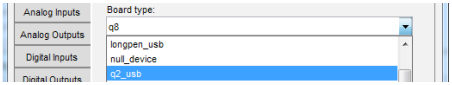
C



Open the Simulink model file (.mdl) found under the Quick Start folder on your hard drive.

D

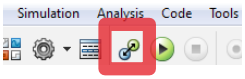
Double-click on the HIL Initialize block and choose the board that is installed on your system (e.g. Q2-USB).



E

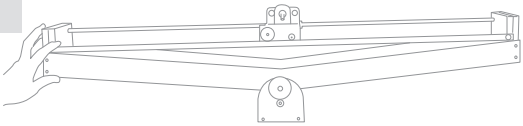
Click on the **Build Model** button on the Simulink model toolbar.

F



Once the model code has been compiled, click on the **Connect To Target** button.

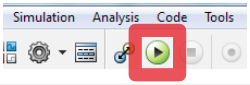
G



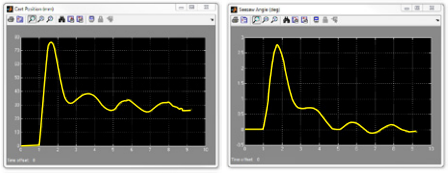
Manually, move the cart to the center of the rack and hold the Seesaw system balanced.

H

Click on the **Run** button to start the QUARC real-time model.



I



Scope view of the VI Front Panel

The cart should be moving back and forth to balance the Seesaw horizontally. The scopes should look similar to those shown here. If not, consult the *Troubleshooting* section at the end of this guide.

J

Click on the Simulink **Stop** button to stop the running model.

TROUBLESHOOTING

Review the following recommendations before contacting Quanser's technical support engineers.

1. Make sure the cables are firmly connected.
2. Check the connection outlined in Step 4 in this guide.
3. Make sure the Linear Servo Base Unit has been set up and tested successfully. Review the Linear Servo Base Unit Quick Start Guide and User Manual setup or troubleshooting section for more information.

Getting an error when trying to build or run the Quick Start Simulink model (.mdl)

A. Type ver in the *Matlab Command Window* and verify that QUARC is on the list. If not, then go through the QUARC Quick Installation Guide to install QUARC. If it is listed, run *mex -setup* as described in the the QUARC Installation Guide.

B. If the "... specific kernel level driver for the specified card could not be found" error is prompted when you attempt to run, then you may not have selected the correct data acquisition (DAQ) device in the HIL Initialize block or the DAQ device has not been installed properly (refer to the DAQ device User Manual).

The Motor is not responding.

A. Review connection Steps 4G and 4H.

B. Ensure the power amplifier is powered on and operational, i.e., when using VoltPAQ-X1, verify that the green LED is lit.

C. Verify that the data acquisition (DAQ) device is functional. Go through the DAQ User Manual for troubleshooting guidelines.

D. See the Linear Servo Base Unit User Manual for more troubleshooting information.

The Encoder is not reading.

A. Review connection in Steps 4I and 4J.

B. Verify that the data acquisition (DAQ) device is functional. Go through the DAQ User Manual for troubleshooting guidelines.

C. See the Linear Servo Base Unit User Manual for more troubleshooting information.

STILL NEED HELP?

For further assistance from a Quanser engineer, contact us at tech@quanser.com or call +1-905-940-3575.

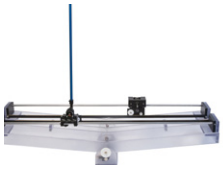
Expand the Linear Servo Base Unit to the following popular experiments using Quanser Linear control add-on modules.

Linear Pendulum

Linear Flexible Joint

Flexible Inverted Pendulum

Seesaw Pendulum



LEARN MORE

To find out about the full range of Quanser control experiments, visit www.quanser.com

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